Python Assignment
Topic-2: Programming Style

Q1. Understanding Python's PEP 8 guidelines.

Ans.

PEP 8 is the official style guide for writing Python programs. It helps programmers write clean and readable code. It gives rules for spacing, indentation, naming, and formatting. By following PEP 8, the code looks neat and easy to understand. It also helps different developers work in the same style. This improves teamwork and reduces confusion. It makes debugging easier because the code is well structured. Beginners can learn proper coding habits using PEP 8. Overall, it is very useful for writing professional Python code.

Example:

```
student_name = "Shivani"
total_marks = 85

print("Name:", student_name)
print("Marks:", total_marks)
```

Q2. Indentation, comments, and naming conventions in Python.

Ans.

Indentation means giving spaces at the beginning of a line. In Python, indentation is very important because it defines the structure of the program. It tells Python which statements belong together. Comments are used to explain the code and are not executed by Python. They help in understanding the logic of the program easily. Naming conventions are rules used to give proper names to variables and functions. Names should be simple, meaningful, and easy to understand. Good naming and proper indentation make the code clean and readable.

Example:

```
if 5 > 2:
   print("5 is greater than 2")
```

Q3. Writing readable and maintainable code.

Ans.

Readable code means code that is easy to understand. Maintainable code means code that can be easily updated in the future. To write good code, we should use simple and meaningful variable names. We should also write comments where needed. Proper indentation should always be followed. Avoid writing too much code in one line. Code should be divided into small parts so that it is easy to manage. Avoid repeating the same code again and again. Clean code helps in easy debugging and future changes.

Example:

```
# Program to calculate total marks
math = 80
science = 85
english = 75

total_marks = math + science + english

print("Total Marks:", total_marks)
```